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Chapter Author(s): SUSAN BROWN and LAURA MANDELL

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Infrastructures

Feminist Infrastructure Building

SUSAN BROWN AND LAURA MANDELL

Our lot is cast with technoscience,
where nothing is so sacred that it cannot
be reengineered and transformed so
as to widen our aperture of freedom,
extending to gender and the human.

—Laboria Cuboniks, “Xenofeminism”

A feminist digital-humanities **infrastructure** is an infrastructure that resists in some way, shape, or form the bifurcation of humanity into m/f. Catherine D’Ignazio and Lauren Klein offer this task of feminist design—and it applies to everything from designing interfaces and visualizations to designing infrastructures—as the first and most important: “Rethink binaries.”¹ First outlining the challenges to thought or critique posed by the tendency of infrastructure toward the naturalization or transparency of infrastructure, this chapter moves from the less to the more visible in pulling out common threads from feminist engagements with infrastructure. Such efforts can be viewed as contributions to “advancing scholarship related to diversity” through what Alan Liu calls a “**diversity stack**”² although we position them as forming a feminist infrastructural ecosystem that helps resist binary or reductive thinking.

*Nail (2016) constantly
comes to mind when I think
of bifurcation as a bordering process*

*digital humanities scholarship seems to
be especially interested in the stack as a
model for technological infrastructure*

But What Is an Infrastructure?

Infrastructure can be characterized as happening invisibly, behind the scenes, for instance to make water flow from a faucet when turned on. As we use them, infrastructures pull our behavior into rivulets of common practice; insofar as cognition is embodied, our bodies absorb knowledge and affect as we turn the dials, which is to say, strike the keys. Infrastructures define affordances, interpellate subjects, make arguments, foster

*Star (1999)
identifies these key characteristics
of infrastructure*

relations and intimacies; and they are themselves in many respects defined both by the subjects who engage with them and their local, institutional, and global conditions of possibility. As Liu notes, infrastructures create “the social-cum-technological milieu that at once enables the fulfillment of human experience and enforces constraints on that experience.”³ But when infrastructures are emerging, as are digital infrastructures, they aren’t yet infrastructures because they are not operating so smoothly as to fade into the background. Their “smoothness” comes not only from the material technology of the infrastructure, but also from adaptations made by the users of the infrastructures: what once required intellectual effort becomes an unconscious raising of the hand to turn on the faucet, made conscious as a gesture only if the faucet is broken or the water too hot.⁴

~ possibly tied to Bourdieu's habitus?

Brown (2016) argues that this information and digital communication is a defining feature of the layer of the stack in particular

Although public infrastructures are large, well organized, and relatively slow-moving, digital infrastructures proliferate quickly around any demand for digital communication and information delivery, which is to say they encompass back-end and front-end information technologies.⁵ In the digital context, multiple contenders for the status of infrastructure arise; some become established through institutions or habits, and others die out or transform into something else. Academic infrastructure may be enveloped in feminized notions of service and embodiment, troubling fundamental category work and distinctions that otherwise go uncontested.⁶ In an ideal process, users and developers work together through give-and-take to develop an infrastructure that isn’t too hard to use habitually—and for which becoming acclimated has enough incentives to make using it worthwhile, anything from gathering status as a super user to accessing a movie or an ebook.

Dunbar & Stor (1999) highlights several examples of users' role in navigating the design, maintenance, and use of early layer infrastructures (i.e. the ICB) This highlights the degree to which infrastructure is itself a design problem, and thus also an ontological or (world-)building issue

This code reminiscent of Brown's (2016) principle of a generative design platform

Digital infrastructures have to be worth it to use—worth it partly thanks to their design, but also partly because the incentives were right and the adaptations relatively easy. Marketing as well as other social forces shape user desire. For example, compare the limited incentives to adopt email in the 1980s to those now: some digital forms cannot be submitted without an email address. If a piece of software isn’t worth it, if users can get what they want more easily another way or stop desiring it because it’s too difficult to access, the digital infrastructure will fall out of use. Infrastructures are nothing apart from use, that is, nothing apart from people.

- further connection to Brown (2016)... the user stack is where the effects of computational infrastructure are personalized

Although infrastructures are both visible and tangible, their use is predicated on their becoming naturalized and incorporated into actions, diminishing users’ ability to discern their influence. Susan Leigh Star points out that infrastructures become “transparent,”⁷ not in the sense that they fully disclose themselves, but rather in the sense that a window is transparent for viewing the world. They are transparent because people do not have

to think about them when they use them. No one has to know how to lay asphalt in order to drive on a highway, but a person could learn a lot about how to lay asphalt if stuck in traffic for road repair.

What it takes to build and sustain infrastructure, as well as how it works, becomes visible when it breaks.⁸ This fact sets up a double-bind for those who want to build equity and/or advocacy into their own systems.⁹ An infrastructure embeds point of view, argument, and values in the way it organizes information and the relationships between its parts, in the kinds of knowledge it both presumes and valorizes, and the affordances it offers to users, whether through the back end or through a graphical user interface (GUI). Infrastructure builders make choices that define the agency of their users. Such choices are always a product of technological affordances and constraints, social relationships, historical context, communities of practice, and paradigms.

~ however users may respond in ways not anticipated by designers

Digital infrastructures channel their specialized worlds. Ideologically laden, their operations become part of ^{reconstituted} habitus, of everyday life,¹⁰ and, as Liu implies, interwoven with ideology in the fabric of lives. Infrastructures mediate conscious thought formulation about tasks into gestures that are unthought, automatic. An early site design book carried the title *Don't Make Me Think*.¹¹ Infrastructures replace thinking with the shorthand of an embodied gesture or the familiarity of an established workflow; enabling a filter on a dating or real estate site replaces with one click the need to reaffirm, repeatedly, certain choices.

~ and the visual nature of GUI and other interfaces is what makes this acquisition of embodied knowledge and gesture possible - see Bratton (2016) on weaponized skeuomorphism

The goal of media critique is to break such processes open in order to bring to light the premises and assumptions, the norms and values, built into an infrastructure, and then, via that knowledge, to disturb its functioning. Builders of infrastructure concerned with equity benefit from critique insofar as it promotes understanding what kinds of things infrastructures naturalize. But here we find the paradox, the double bind: The power of infrastructures lies not only in *what* they naturalize, but *that* they naturalize. If we were to practice "critical making" by building an infrastructure that refused to replace conscious thought with filters, that refused to obscure the decision-making process in each one of a series of repetitive gestures, people would stop using it. They might even say it was broken. If "Don't Make Me Think" is the mantra of existing infrastructures, how can we offer alternatives that are desirable while thought-provoking?

To think about feminist infrastructure building, we begin with less visible considerations of labor and standards, followed by hardware and software, and finish with interfaces, the most visible aspect of digital infrastructures. In what follows we track research and practice that address features such as obfuscation and simplification in different layers of what is increasingly

~ this is the layer that mediates the connection between user and the rest of the planetary infrastructure of computation and the stack

Feminist Infrastructure Building

called the “stack.” Feminist infrastructure initiatives, better understood as contributing to an ecosystem rather than a stack, resist or rethink the sexism imbricated in systems.

Critiquing the Stack

Given that infrastructure “has much of the same scale, complexity, and general cultural impact as the idea of ‘culture’ itself,”¹² we are in the midst of a more general turn toward understanding infrastructure as a broadly constitutive force. Benjamin Bratton’s (densely theorized) and (seductive) description of a planetary megastructure that overwrites traditional understandings of geopolitical sovereignty employs the metaphor of “The Stack.” His complex and evocative model attempts to move beyond binaries¹³ and has been adapted by Liu to articulate the powerful notion of a “diversity stack” for Digital Humanities (DH).¹⁴ Yet, although Bratton’s notion of geodesign “draws from (and into) its specific planetary situatedness,” and he positions “Earth” as the Stack’s “schematic foundation. . . the driving force and form of its logic,”¹⁵ *The Stack* emphasizes the rigidity and stability of the “technological totality.”¹⁶ DH may be considered a small component of such a “stack” and as such to reflect its characteristics, for instance in the four layers of the open-source LAMP stack—the Linux operating system, Apache server, MySQL database, and PHP, Perl, and Python code often employed to build websites.¹⁷ The feminist engagements with infrastructure we explore here offer an alternative emphasis on feminist infrastructural contributions as constituting not a stack but an ecosystem situated not only within academia but also in particular contexts, with an accompanying interest in the forms of relationality, resistance, and dynamism that are occluded by Bratton’s stress on governance.

Feminisms offer a wide range of generative and critical engagements with infrastructure building for “tracing circuits of power, norms, and agency that are realized in particular modes of relations.”¹⁸ Without falling into facile charges of DH as neoliberal, we recognize with Miriam Posner that the technologies we use entail impediments to creating infrastructure that embodies radical critique. Infrastructures impart normative values, “values and ethical principles [that are] inscribe[d] in the inner depths of the built information environment”¹⁹ but, despite their universalizing drive, infrastructures are always already operating within and shaped by particular social, institutional, and national contexts. Infrastructure building on feminist principles thus offers a powerful form of critique while grappling with myriad constraints and determinants.

infrastructures are community based, after all, as Star (1999) argues

Labor

Historians of infrastructures such as the railroad and the telegraph point to the pain and duress inflicted upon the laborers forced to build them.²⁰ For academic institutions currently confronting new technologies, labor causes duress in the absence of supportive institutional infrastructures for DH work; precarious labor is a matter of serious concern in these efforts.²¹ We focus here on the labor of building specifically digital infrastructures, which is caught up in this larger concern since such infrastructures, particularly feminist ones that are funded from grants, are often built and maintained by teams partially or fully staffed with precarious labor. As the technological background, infrastructure is “a system of substrates”²² that is invisible unless it breaks, planting it firmly in the (re)productive economies of housework and carework.²³ Its functioning model is therefore gendered as feminine, along with all the pitfalls that entails: the invisibility of labor and the failure to credit this work as both an intellectual and laborious endeavor. Debates over infrastructure often elide the essential human agents without which no infrastructure can run. The elision of human contributions sustains the illusion of infrastructures as neutral and equally accessible, making the importance of labor-centric initiatives such as the FemTechNet or HASTAC networks hard to perceive as infrastructure.

Matters of care,
repair, and
maintenance

The gendered association of infrastructure with service²⁴ has inflected the rise of concerns around **care and repair**.²⁵ Infrastructure work is not associated with individual credit; it is unsexy intellectually to the extent that it creates the pipes that other people send stuff along or sustains the conditions for intellectual labor. Like reproductive and domestic work, the invisible repetitive labor of creating and sustaining academic infrastructure falls outside of dominant models of productivity, typically located beyond the funding structures that sustain research. For example, in Europe, major DH infrastructure initiatives like CLARIN and DARIAH have been funded separately from research, whereas infrastructure created by smaller research projects often has great difficulty sustaining itself. This can create challenges when there is a close relationship between infrastructure building and feminist critique. This is not to say that such models cannot contribute to feminist infrastructure efforts. The Humanities Networked Infrastructure (HuNI) was funded as part of a large infrastructure funding campaign in Australia,²⁶ as is the Linked Infrastructure for Networked Cultural Scholarship (LINC). Funding models that divorce research from infrastructure, however, work against **critical infrastructure development**. A refusal to fund research time per se in connection with infrastructure projects reinforces the view that

it's important to note that the employment of the m/g binocular distinction is a distinctly holistic infrastructure practice

As is especially observable in digital infrastructures, standards subtend infrastructure insofar as they require widespread implementation. In so doing, they homogenize. Advocates of global DH and critical race theorists advocate resisting such homogenization as oppressive. For example, there is the movement for **minimal computing** as advocated by many in the Global South or GO::DH, or the “poor theory” of Ackbar Abbas and David Theo Goldberg, whose “commitment to open source input and incessant reformulation entails that it values heterogeneity over homogenization, intensely interactive pluriculture over monoculture, crowdsourcing over hermeticism, with constant recalibration in the face of error, failure, and loss.”²⁹

lots of
resonance
with
Arturo
Escobar's
pluriverse

ideological
(un)marking

A commitment to the critical use of technology within DH can be seen in feminist initiatives to establish, apply, and maintain standards. The Text Encoding Initiative (TEI) has grown through a symbiotic relationship with the “encoded bodies” of the Women Writers Project;³² the metadata format of the Advanced Research Consortium, inherited from NINES.org, adopted the Resource Description Framework to promote contextualizing, remixing, finding things through terms in current scholarly use, and sharing through a community folksonomy;³³ and the ontology of the Canadian Writing Research Collaboratory represents the intersectionality and contingency of social identities.³⁴ Data models and standards are infrastructural resources that have pervasive impacts across a wide range of systems and contexts. For instance, Tara McPherson found that data models supported by relational database technologies were crucial in enabling designers and researchers collaboratively to “create scholarship that could not exist in print” through the journal *Vectors*.³⁵ Likewise, the TEI implementation of the Extensible Markup Language (XML) standard provides a digital model or “theory of the text” that has been applied to the creation of “digital editions, teaching anthologies, research corpora, full-text thematic research collections, digital archives,” as well as derivative datasets and presentation modes such as graphs, maps, and network visualizations.³⁶

Hardware

Still most opaque in some ways is the most fundamental level of infrastructure, the material layer of hardware that many would equate with infrastructure. There are so many layers of abstraction between the movement of electrons through circuit boards and the operations of a computer that their implications are hard to grasp, but many feminist theorists have resisted facile mappings of either gender or mind/body distinctions onto binary code.³⁷ Hence, in Donna Haraway’s figure of the **cyborg**, technology is incorporated in the body as infrastructure.³⁸ Historicizing our relationship to the materiality of computing is key. Lori Emerson promotes “infrastructural thinking” through undertaking media archaeology work from a situated and embodied perspective.³⁹ Both by working with original machines and looking at industry debates over interfaces, she traces how we have arrived in a time when “users have little or no comprehension of the digital computer as a medium.”⁴⁰

—have the increasing embeddedness of infrastructure and its materiality by its increasing distance from its materiality

Networks are both a powerful metaphor and realities that transmit information or knowledge in the form of power. They enable, but also provide government and corporate entities with the means of surveillance and extraction, as well as of controlling access. Feminist infrastructure, such as the **artivism** of micha cárdenas discussed below, may require autonomy

from or subversion of global networks, and inequality of network access may mean infrastructure must be accessible offline or adaptable to sporadic connectivity, as highlighted by the minimal computing movement. Rather than viewing the net as a hegemonic totality, Emerson argues that understanding the implications of networks on which much DH infrastructure runs means unpacking the messy, collaborative, processual, and contingent histories obscured by “Great Man Stories.”⁴¹ *the ideological work and power of narrative and canonization*

Feminist critical making involving wearable computing elucidates and intervenes in our embodied relationships to digital devices.⁴² Kim Brilliant Knight highlights the intersection of accessibility and transparency in the LilyPad Arduino designed for wearable apps: “Leah Buechley, the creator of the LilyPad, describes it as having a soft, colorful, beautiful aesthetic that she hopes will affect the world of engineering applications. As a result, both the LilyPad controller and the stitches used to connect it to other components are often consciously made visible as part of wearable projects.”⁴³ Because LilyPad projects are often fabric- or fashion-based projects, they trouble the boundaries between the gendered arenas of “high” tech as opposed to lowlier “craft” activities oriented toward aesthetics and play; they shift the materiality of computing infrastructure from industrial to domestic contexts.

Nevertheless, computational hardware is a profound factor in infrastructure. Whereas Bratton argues that the stack is not deterministic, McPherson has argued that the very basis of contemporary computer operating systems is founded on a “lenticular” logic in the way that certain things are partitioned off from others as computers store code separately from data.⁴⁴

Accordingly, despite her ground-breaking work on the open-access Scalar platform, McPherson says “I know that Scalar is not a deep challenge to the workings of corporate and computational capital.”⁴⁵ The admission acknowledges the major impediments to imbuing infrastructure with the critical values to which one adheres. The technologies that undergird publishing platforms or virtual research environments impose real constraints on how systems can be built, particularly if they are to meet accessibility requirements, work modularly, and adopt standards—accepted data formats and established technologies.

Software

Much explicitly feminist infrastructural work has involved software in one way or another, perhaps because it is more amenable to intervention than hardware. The set of programs, procedures, and routines that tell a computer what to do, software is most visible to users in playing digital games. Digital artist Mary Flanagan teaches critical game design from a feminist perspective. Not only is she a great maker, as can be seen in her 2009 book *Critical*

Play: Radical Game Design, but she is also critically adept at unearthing the values imparted by digital games.⁴⁶ Jane McGonigal, another game designer, melded self-help discourse to gamification in *Superbetter*, an app designed to counter her own struggles with post-concussion syndrome.⁴⁷ Zoë Quinn used the Twine platform to create the deceptively simple game *Depression Quest* (2013), which became the target of GamerGate, as an educational tool. Such gaming projects understand software not as an escape, but as an intervention in the material world.

Infrastructure does not emerge in isolation, Lucy Suchman argues. Its profound imbrication with lives and subjectivities, bodies and built worlds, therefore situates it as a prime location for critical intervention along with other forms of knowledge organization. McPherson, a self-described “feminist in a software lab,” designed *Vectors* and *Scalar* for difference, to create an intersectional, collaborative space within which scholars and users could “rethink their relationship to the computational and to enter into a complex assemblage of aesthetics, technology, pedagogy, creativity, and new publics.”⁴⁸ Anne Balsamo draws on the work of Karen Barad to stress the extent to which iterative design and development constitute a situated form of inquiry into the complex and shifting “intra-actions” among subjects and their environments, an inescapably relational process in which the diversity of participants matters.⁴⁹ For instance, the software created for an “interactive memorial,” digital experiences constructed around the AIDS quilt, includes: (1) an open-source mobile web application called AIDS QUILT TOUCH; (2) a tangible tabletop interactive that enables viewers to SEARCH the database of Quilt images to find a specific image and to BROWSE the archive of Quilt panel images; and (3) a community sourcing application that engages people in analyzing and archiving information about the Quilt.⁵⁰

Collectively, this software counters the official historical narrative about interactions between AIDS “patients” or “victims” and the “hero scientist”⁵¹ with a less homophobic history of intra-actions, debunking the notion “that there are separate individual agencies,” in this case, human actors, and replacing it with the idea that the causal agents mitigating the AIDS epidemic “emerge[d] through” relations among all the materialities, power dynamics, individuals, and groups involved.⁵² This perspective resists an understanding of software as a stable thing “run” by a separate person, and instead sees agency as more complexly dispersed across a range of human and nonhuman interrelationships: “the digital application was never a singular thing with set boundaries or well-defined edges. It was always, and still is, an unfolding set of possibilities, animated in the intra-actions that include human fingers striking keys, middleware reading machine code, machine code acting on lines of zeros and ones, materials conducting energy in the form of heat, and so on.”⁵³

Bennett (2010)
points to this
notion of
agency as
distributed

Relations and Intimacies

Deb Verhoeven points out that attending to the “acts and technologies of ‘connection’” will “move us beyond thinking of connectivity as binary or thinking of infrastructure as a form of mediation” to “address the social or relational aspects of infrastructure.” For her, “the restless, transformative, and connective work of infrastructure can be understood as a form of inventiveness and interpretive resourcefulness,”⁵⁴ manifested in the HuNI infrastructure that enables people to create networks by defining links between objects across different cultural collections rather than working from a fixed taxonomy or ontology.

In contrast, the Advanced Research Consortium (ARC), led by Laura Mandell and built on information architecture developed by Bethany Nowviskie,⁵⁵ *does* require participants to adhere to multiple intersecting taxonomies. However, ARC metadata categories have been created from the ground up as it grew from NINES.org to 18thConnect.org to mesa-medieval.org to modnets.org and studiesinradicalism.org and beyond.⁵⁶ Each new organization’s digital projects need new metadata categories for discoverability, which are then debated and decided on by all directors of the ARC communities.⁵⁷ ARC metadata thus evolves to meet the needs of the larger scholarly community, and its “agile metadata development”⁵⁸—agile like the tortoise—enables changes when exclusions are identified. Although they take apparently contradictory approaches to metadata development, both HuNI and ARC are grounded in the principle that taxonomic infrastructures must emerge from and be responsive to the communities they serve.

Such relations engender intimacies. As Ara Wilson puts it, “Infrastructure is a systematic assemblage of objects, codes, and procedures that, whether it fails or succeeds, is often an embedding environment for intimate life.”⁵⁹ As lived components of daily lives, infrastructures “mix human and nonhuman agencies” and “are entwined with social relations.”⁶⁰ Jacqueline Wernimont therefore calls attention to **interface** as relationship: “[D]igital archives are [not only thresholds to things but] also thresholds between actions. That ‘thingness’ and those actions are as much an experience of the user as they are of the encoder, programmer, and editor. Finding ways to enable user engagement in production would . . . embody a more radical feminist approach to our understanding of technology as entailing ‘interplay between designing and use, or between designer and user’ [Rosser 2005, 11].”⁶¹

relationship
between
design and
use

The logical extension would be dynamic interfaces in which users could partner more deeply and radically than in the past not only in the contribution of content, which is the main concern in Wernimont’s piece, but also in the ongoing reshaping of the infrastructures that in turn shape them.

Interface and Access

Reflecting the profound regard for relationality that runs through intersectional feminist theory, Milena Radzikowska and collaborators articulate six principles for **feminist interface design**:

1. Challenge existing methods, beliefs, systems, and processes;
2. Focus on an actionable ideal future;
3. Look for what has been made invisible or underrepresented;
4. Consider the micro, meso, and macro;
5. Privilege transparency and accountability; and
6. Expect and welcome being subjected to rigorous critique.⁶²

These principles notably stress intention and analysis rather than full instantiation of them, since limitations ranging from resistance in the technological materials⁶³ to funding constraints work against the full embodiment of such goals within a production-quality interface. In addition, developing new interface languages and affordances based on such critique requires iteration and community participation, from published reviews of digital work to more effective user-designer feedback loops, requires time and attention. However, the recent flowering of **feminist design thinking**, such as that proposed by Catherine D'Ignazio and Lauren Klein,⁶⁴ bodes well for future initiatives.

Although not necessarily engaging directly with the **rhetoric of interfaces**, much feminist infrastructure work focuses on access to digital content and spaces. One area of broad activity is in content creation that enables access to otherwise unavailable material or perspectives. As cultural analytics become more influential, the provision of well curated and structured datasets becomes increasingly pressing. As many have pointed out, our research is only as good as our data, and much of our data is woefully impoverished, either because it is mute on the subject of gender and other identity categories, or because it is biased by the processes of collection, often of the libraries or archives on which it is based, so as to underrepresent women, racialized, 2SLGBTQIA+, and other marginalized authors or content.⁶⁵ By analogy with the library as the so-called "lab" or fundamental infrastructure of the humanities, the production and open dissemination of digital content, no matter how modest, that enables feminist analysis and reuse becomes a component of an ecosystem of feminist infrastructure. As experiments in the conditions of possibility for feminist work in DH, multitudinous projects have been structured to facilitate inquiry into gender and other intersecting social categories. These projects now comprise a vital component of our research infrastructure. But binaries threaten to solidify into essential or biological categories in the field of text-mining, or **"cultural analytics."**⁶⁶

issues in data phenomena and the rhetorical power of data collection, cleaning, analysis, and visualization

Making content accessible in ways that offer alternatives to binaries and promote reuse itself requires infrastructure, of course. Many librarians have embraced open-access publication as a form of social justice. The Canadian Writing Research Collaboratory (CWRC) emerged from the Orlando Project as a space in which other projects could create, publish, and share content.⁶⁷ CWRC's online tools also aim to eliminate often gendered barriers to participating in DH by reducing up-front costs and the level of technical expertise required. Marii Nyröp's Wax software for producing static exhibition websites⁶⁸ emerges from the minimal computing movement, which recognizes that many are disadvantaged because of global inequalities by uneven access to expertise, hardware, and bandwidth. Wax meshes beautifully with a feminist concern for the gendered dynamics of care and repair. Such infrastructures allow disadvantaged groups to mobilize content and simultaneously promote standards and interoperability for reuse and preservation, so that open-access content can have a future.

Open access can in certain contexts pose as much of a problem as copyright, however.⁶⁹ Privileging openness can run against the need of some communities for control or privacy, given how their cultural knowledge has been expropriated in the past. Leveraging "Indigenous models of information management," Kimberly Christen says, the Mukurtu content management system "was built from the ground up with Indigenous communities in mind."⁷⁰ A central feature of Mukurtu is the ability to restrict access to members of particular communities, and to apply Traditional Knowledge Labels to indicate terms of local access and use, which may for instance be seasonal or gendered. Some libraries are now beginning to incorporate Traditional Knowledge licenses in their cataloging.⁷¹

The "open" web and many social media platforms in fact provide a virtual playground on which people, mostly men, engage in very real harassment of and violence against women, people of color, queer and trans people, Indigenous people, and other marginalized groups with the effect of making digital spaces less accessible, as in the case of GamerGate.⁷² Hence the importance of private spaces within which groups can engage in networking, education, and research. FemTechNet, for instance, has operated both as an open and as a closed (but still welcoming) group using a range of social media platforms and tools,⁷³ illustrating the extent to which feminist infrastructure can in some cases be defined by a group's relational intentionality and community practices rather than specific technologies.

Physical spaces are also crucial. Many DH centers led by women, along with the emergent feminist maker-space movement,⁷⁴ are all about making technological spaces safe, accessible, and welcoming to women. Wernimont and Losh invoke the exemplary "Local Autonomy Networks (Autonets)"—off the corporate internet grid, communicating among members via

risks of
(infrastructural)
visibility and
access

wearable garments—designed by artist micha cárdenas.⁷⁵ cárdenas sought to create these networks of communication as “abolitionist infrastructures for community-based safety”⁷⁶ to be used to resist violence against women, 2SLGBTQIA+ people, people of color, and other targeted groups. Although technological limitations led to a shift in focus toward embodied gesture and performance, the project underscores the extent to which technological affordances emerge within or are refused by particular spaces. All of these projects evince an **ethics of care** like that enacted in the summer of 2020, when the COVID-19 pandemic made it impossible to hold the international Digital Humanities conference as planned in Ottawa. Pivoting to an online conference for more than nine hundred registrants was facilitated by the open, inclusive, and collaboratively produced infrastructure of the Humanities Commons WordPress customization, directed by Kathleen Fitzpatrick and an instance of the infrastructural conditions in which the “generous thinking” Fitzpatrick advocates becomes real.⁷⁷

A Feminist Infrastructural Ecosystem

Verhoeven renames feminist infrastructure “**infrapuncture**,” gesturing toward acupuncture or relieving stresses. She invites us to “think about what a better version of infrastructure could be that could intervene at critical locations.” She reminds us that infrastructures are “circuits of action and reaction” that “do not begin and end with technology,” and that building an infrastructure requires “a theory of world making, if we are to matter.”⁷⁸

Feminist DH infrastructure work is best framed theoretically, we have argued, not as a stack, but as an ecosystem in which actors and activities interrelate and intra-act, impacting each other in the context of local environments that themselves differ remarkably. It encompasses the provision of training initiatives, spaces, and networking opportunities enabled by conference organization and digital humanities centers, as well as the installation and running of hardware or the creation of software designed in some way to recognize, mobilize, or embody the difference that gender analysis makes. It also includes content-provision projects that might not seem, on their face, to constitute infrastructure. A feminist DH infrastructure aims to redistribute power in order to achieve equity and empowerment.⁷⁹ It consequently embraces pluralism: identity, inclusivity, and diversity. It respects all participants as agents, giving credit where it is due and making labor visible.

Suchman describes the kind of responsibility involved in infrastructure:

Agencies—and associated accountabilities—reside neither in us nor in our artifacts but in our intra-actions. The question, following Barad, is how to configure assemblages in such a way that we can intra-act

—feminist infrastructure is that pluralised in nature

I'm still unsure of this - I'm not convinced this is how Suchman's (2016) notion of the stack is intended to be taken up, though they may be made directly in response to the use of the (diversity) stack as a (visible) model of the digital humanities

responsibly and generatively with and through them. [. . .] [The] perspective suggested here takes persons and machines as contingently stabilized through particular, more and less durable, arrangements whose reiteration and/or reconfiguration is the cultural and political project of design in which we are all continuously implicated. Responsibility on this view is met neither through control nor abdication but in ongoing practical, critical, and generative acts of engagement.⁸⁰

goals and
defining
characteristics
of feminist
infrastructure
for the
digital humanities

The “ongoing practical, critical, and generative acts of engagement” that constitute contributions to feminist DH infrastructure, broadly conceived, involve rethinking binaries, making labor visible, embracing pluralism, acknowledging affective and rhetorical effects, and redistributing power. An infrastructure becomes “feminist” by making apparent in some way through its own inclusiveness the unstated exclusions endemic to digital infrastructures. Although infrastructure building fueled by a concern to address gender and other forms of inequity encompasses a broad range of activities and scales from the local to the transnational, from the minimal to the platform, these threads are woven, variously in response to local situations or particular concerns, through them all.

Infrastructures are necessarily constrained both by their situatedness and by the double-bind of instigating change within contexts in which most users expect habituation rather than disruption, seamlessness rather than resistance. But Barad’s generative interpretation of constraints as “dynamics of possibility”⁸¹ invites us to engage in feminist infrastructure work that is animated by critique and care, theory and generosity, and thus advance the project, in the words of Cuboniks invoked at the outset, to “widen our aperture of freedom.”

Notes

Many thanks to Liz Losh, Ravynn Stringfield, and the editors for suggestions that made this a better piece.

1. D’Ignazio and Klein, “Feminist Data Visualization,” §3.1.
2. Liu, “Toward a Diversity Stack,” 130.
3. Liu, “Drafts for Against the Cultural Singularity (book in progress).” For a discussion of definitions of infrastructure from the beginning of the term’s origin, see A. Wilson, “The Infrastructure of Intimacy” (264–65).
4. Bowker and Star, “How to Infrastructure,” 231–33.
5. S. Brown, “Delivery Service,” 2019.
6. S. Brown, “Delivery Service,” 2019.
7. Star and Ruhleder, “Steps toward an Ecology of Infrastructure”; Star, “An Ethnography of Infrastructure,” 381; Bowker and Star, “How to Infrastructure,” 231.

8. Losh, "The Rhetoric of Infrastructure," 22; Liu and Gold, "Session Description—Critical Infrastructure Studies"; Chun, "What We Talk about When We Talk about DH."

9. The field of value-sensitive design argues that values are embedded in computer systems of all kinds and proposes consciously incorporating values rather than taking them as they come. See Hoven, "ICT and Value Sensitive Design"; Friedman, Kahn, and Borning, "Value-Sensitive Design and Information Systems."

10. Chun, *Updating to Remain the Same*.

11. Krug, *Don't Make Me Think*.

12. Liu, "Drafts for *Against the Cultural Singularity*."

13. Bratton, *The Stack*, 355.

14. Liu, "Toward a Diversity Stack," 135.

15. Bratton, *The Stack*, 107, 355.

16. Bratton, *The Stack*, xviii.

17. "LAMP (software bundle)."

18. A. Wilson, "The Infrastructure of Intimacy," 253.

19. Star, "An Ethnography of Infrastructure," 379.

20. Losh, "The Rhetoric of Infrastructure," 2019.

21. Berens, "DH Adjuncts."

22. Star, "An Ethnography of Infrastructure," 380.

23. For a discussion of carework, particularly within a Black feminist graduate student context, see Stringfield's chapter in this volume.

24. Brown, "Delivery Service."

25. Klein, "The Carework and Codework of the Digital Humanities." See also "Forum: Ethics, Theories, and Practices of Care," 423–51.

26. Verhoeven, "As Luck Would Have It."

27. Verhoeven, "As Luck Would Have It," 21. Linked Open Data is machine legible through published ontologies and vocabularies: LINC's datasets can specify gender differently than m/f/other: <https://vocab.lincsproject.ca/>. The project's data structures and vocabularies work together to represent gender and sexuality from an intersectional feminist perspective. See also *Homosaurus*.

28. See Stringfield's chapter in this volume.

29. Goldberg, "The Afterlife of the Humanities." See also Thorat's chapter in this volume.

30. Beneschott, "Is Open Source Open to Women?"; Reagle, "Free as in Sexist?"; "Open Source."

31. Wajcman, *Feminism Confronts Technology*; Hill, Corbett, and St. Rose, *Why So Few?*; Perez, *Invisible Women*.

32. Flanders, "The Body Encoded."

33. Nowviskie, "Collex," 1.

34. Brown et al., "Cultural (Re-)formations."

35. McPherson, *Feminist in a Software Lab*, 118, 128.

36. Flanders, Bauman, and Connell, "Text Encoding," 105–6.

37. E.g., Halberstam, "Automating Gender"; Wajcman, *Feminism Confronts Technology*.

38. Haraway, "A Manifesto for Cyborgs."
39. Emerson, "The Media Archaeology Lab as Platform for Undoing and Reimagining Media History," 175.
40. Emerson, *Reading Writing Interfaces*, xviii–xix.
41. Emerson, "Writing Telematics//Other Networks in Canada."
42. Wernimont and Losh, "Wear and Care Feminisms at a Long Maker Table."
43. Knight, "Making Space," 2017.
44. McPherson, "Why Are the Digital Humanities So White?," 144.
45. McPherson, *Feminist in a Software Lab*, 241.
46. Flanagan and Nissenbaum, *Values at Play in Visual Games*.
47. McGonigal, *Reality Is Broken*, 2011.
48. McPherson, *Feminist in a Software Lab*, 243.
49. Balsamo, *Designing Culture*, 35–36.
50. Balsamo, "The AIDS Memorial Quilt."
51. Wilson, "AIDS Memorial Quilt Digital Experiences Project."
52. Barad, *Meeting the Universe Halfway*, 33. In concrete terms, those who style Dr. Anthony Fauci as the hero scientist combatting the AIDS virus forget, as he acknowledges, that he was dragged kicking and screaming into the research arena by AIDS activists: neither alone created effective treatments; the "hero" of this story are the agencies comprised of their "mutual entanglement."
53. Balsamo, *Designing Culture*, 47.
54. Verhoeven, "As Luck Would Have It," 9, 11.
55. COLLEX—the COLLEction and EXhibit builder—was created for NINES.org (Networked Infrastructure for Nineteenth-Century Scholarship) by Dr. Nowvieskie working with programmer Erik Hatcher, among others; Elizabeth Sadler later worked with Erik Hatcher to create Blacklight, an evolution of COLLEX (Nowvieskie, Sadler, and Hatcher, "Adapting an Open-Source Scholarly Web 2.0 System for Findability in Library Collections"; see also Nowvieskie, "COLLEX").
56. ARC metadata categories are specified here: http://wiki.collex.org/index.php/Submitting_RDF. For more information about ARC as a community, see ARC.
57. Mandell, "Principles for Metadata Reform."
58. Mandell, "Revitalizing the ARC Infrastructure through Linked Open Data."
59. A. Wilson, "The Infrastructure of Intimacy," 274.
60. A. Wilson, "The Infrastructure of Intimacy," 261–62.
61. Wernimont, "Whence Feminism?," para. 23.
62. Radzikowska et al., "A Speculative Feminist Approach to Design Project Management," 101.
63. Nowvieskie, "Resistance in the Materials."
64. D'Ignazio and Klein, "Feminist Data Visualization."
65. S. Brown, "Categorically Provisional"; Mandell, "Gendering Digital Literary History."
66. Mandell, "Gender and Cultural Analytics."
67. S. Brown, "Scaling Up Collaboration Online."

68. Nyröb and Gil, "Credits."
69. It is beyond the purview of this chapter to discuss the open access movement in general, but access to digital archives requires that a number of different infrastructures be in place, legal as well as technological.
70. Christen, "Tribal Archives, Traditional Knowledge, and Local Contexts," 5.
71. "Traditional Knowledge (TK) Labels."
72. Mortensen, "Anger, Fear, and Games."
73. Wernimont and Losh, "Wear and Care Feminisms at a Long Maker Table."
74. Toupin, "Feminist Hackerspaces"; Martin, "Centering Gender."
75. Wernimont and Losh, "Wear and Care Feminisms at a Long Maker Table,"
104. See also the chapter by Wernimont and Stevens in this collection.
76. cárdenas, *Poetic Operations*, 49.
77. Fitzpatrick, *Generous Thinking*, 232–35.
78. Verhoeven, "Identifying the Point of It All."
79. Bardzell, "Feminist HCI," 1301; Fiesler, Morrison, and Bruckman. "An Archive of Their Own," 2; D'Ignazio and Klein, "Feminist Data Visualization," 2–3.
80. Suchman, *Human-Machine Reconfigurations*, 285–86.
81. Barad, *Meeting the Universe Halfway*, 177.

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